

GEOG 281A

Gerrymandering

Do voters choose their representatives, or do representatives choose their voters?

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THE BASICS

Gerrymandering

Gerrymandering is the intentional manipulation of electoral district boundaries to favor or disadvantage a particular group (e.g., a political party, a racial, ethnic group) thereby distorting the translation of votes into representation.

Gerrymandering Basics

- With each decennial census, political districts are redrawn to reflect shifts in population geography. **One person, one vote**
- **Apportionment** (seats to states), **Redistricting** (voters to districts)
- Who draws districts – state legislatures, independent commissions, courts

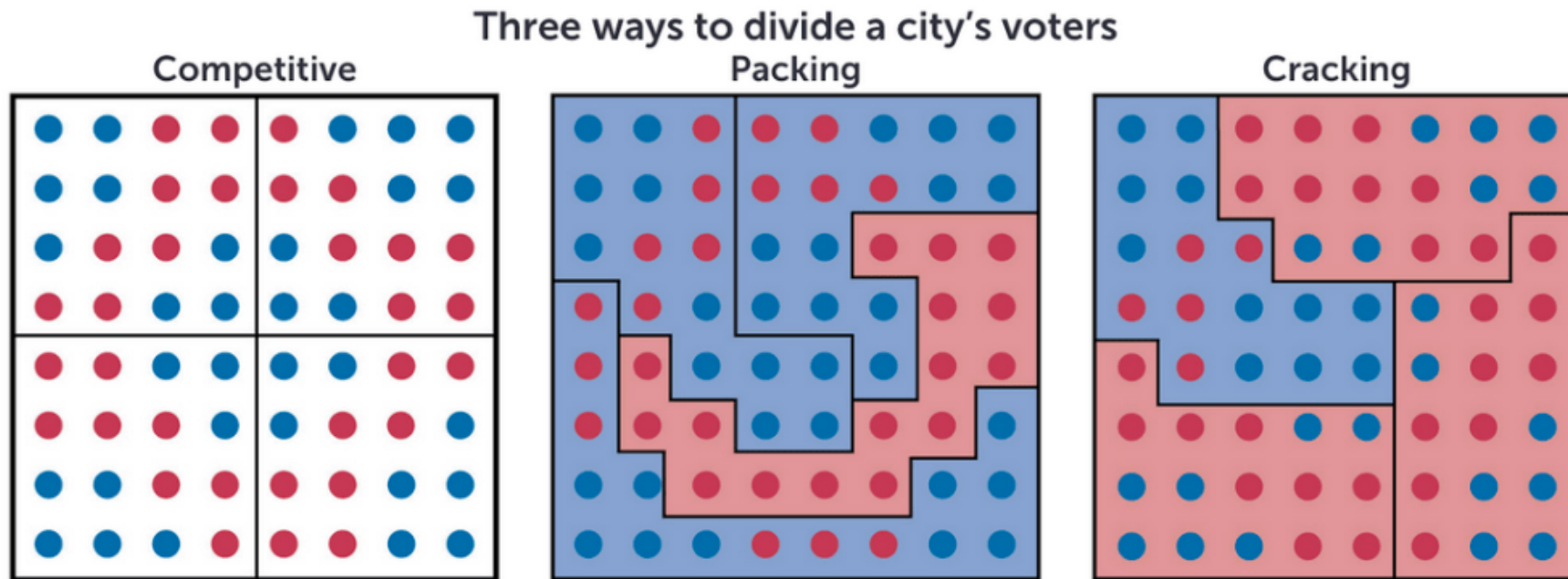
Gerrymandering Process

- Districts are assembled from smaller administrative units (e.g., block groups) or voting precincts for easier election administration
- **Rules:** Equal population, contiguity, compactness, respect for existing boundaries (counties), communities of interest, *Voting Rights Act*

*Many maps will meet the rules, but produce very different political or racial outcomes. **Who draws, or adjudicates, the maps matters?***

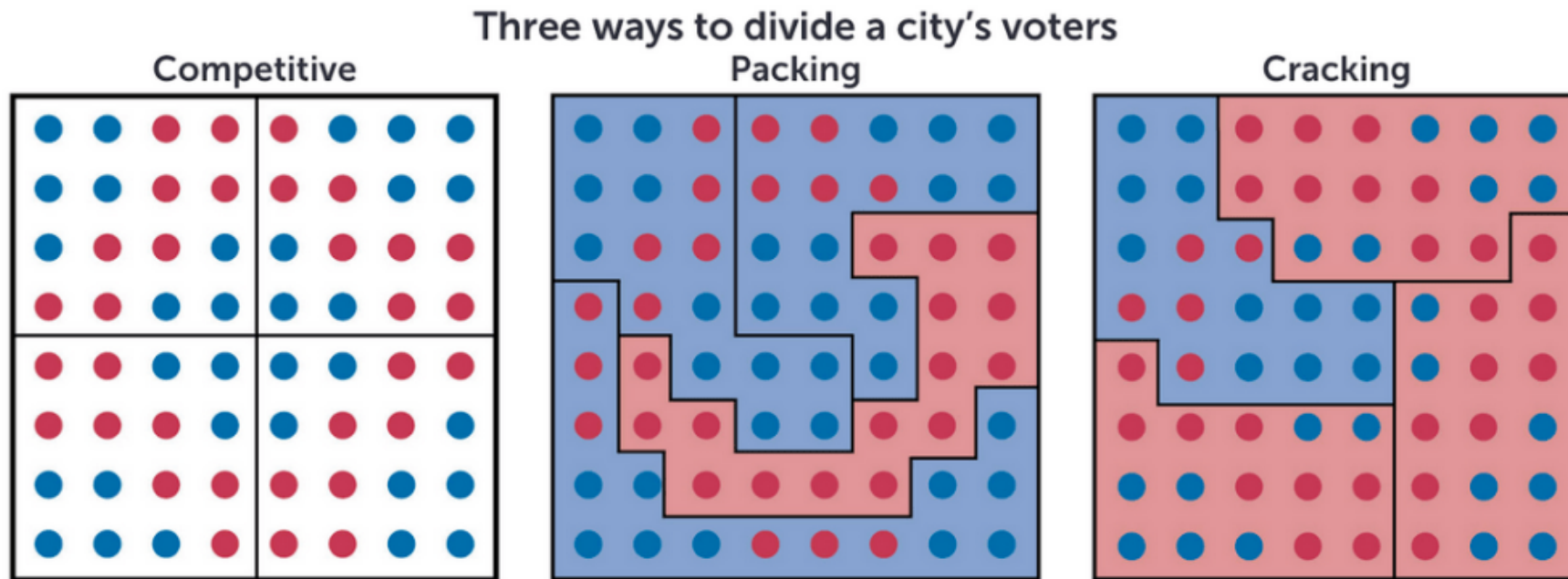
Cracking

Divide a group (political, racial) into multiple districts, so that groups are not a majority in any district.



Packing

Concentrate a group (political, racial) into a few districts, so that group is not a majority in other districts.

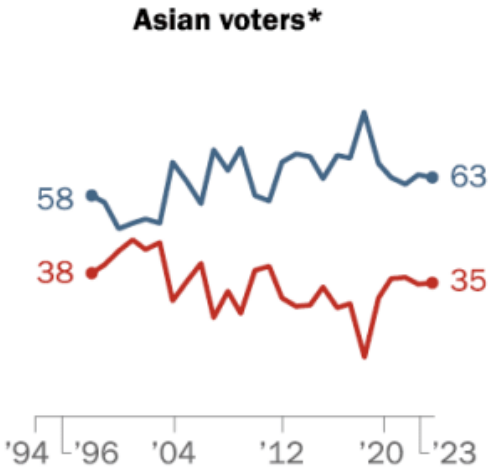
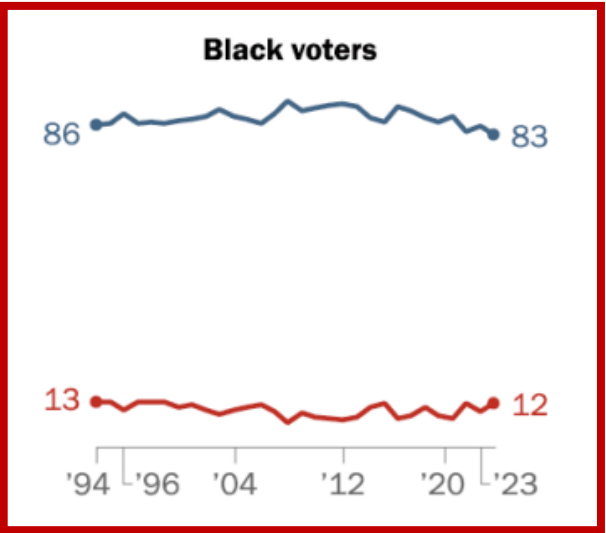
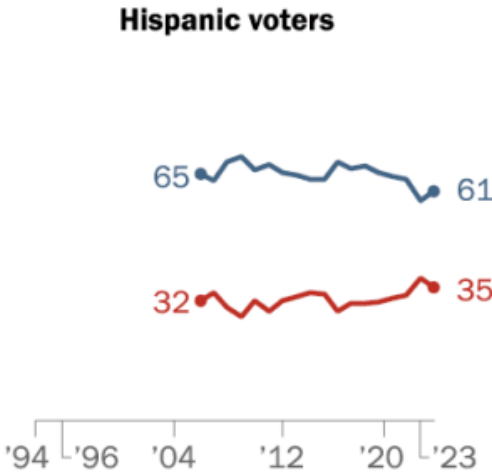
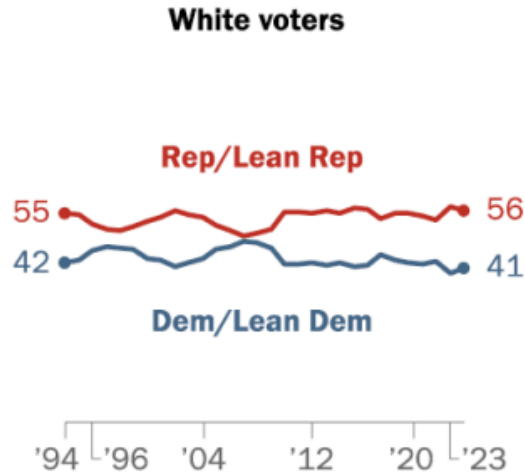


STUDENTS PERSPECTIVE

The scale effect, size, is certainly crucial in this context. If a district is too large, a dense urban minority group gets completely swallowed up and outvoted by the surrounding rural areas. If it is too small, communities get fractured.

**Majorities of Hispanic, Black and Asian voters
continue to favor the Democratic Party, while White
voters remain more aligned with GOP**

% of registered voters in each group who are ...



Notice any
correlation here?

Quick Legal History

Voting Rights Act (1965) law against racial vote dilution given force by the *Thornburg v. Gingles* (1986) test that allows a large, geographically compact minority group claim entitlement to an opportunity district.

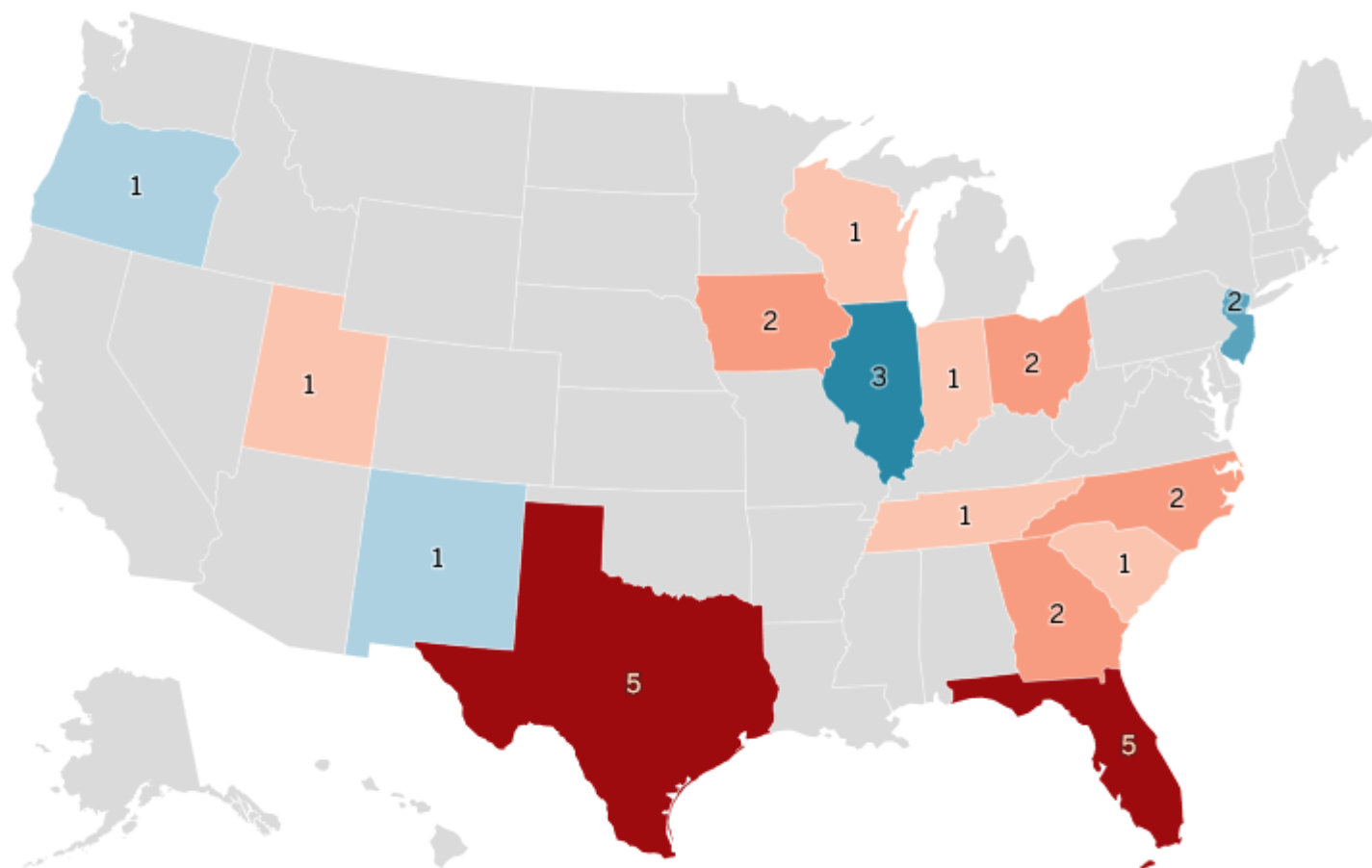
Shaw v. Reno (1993) & Bush v. Vera (1996) identify odd shaped districts as a signal of possible unconstitutional racial gerrymandering.

Rucho v. Common Cause (2019) Gerrymandering is a political question and federal courts cannot review such allegations.

Louisiana v. Callais (2026) As long as it is not the intent to racially discriminate in redistricting — for example, if a map is drawn for partisan gain — then the racially discriminatory consequences of such a map are permissible.

Current Congressional Maps Have a Net 16 Fewer Biden Districts Than Non-Gerrymandered Maps

Extra Seats by Party Using 2020 Presidential Election as Proxy



Source: Brennan Center analysis of Harvard University ALARM Project simulated redistricting plans that comply with anti-gerrymandering standards in the Freedom to Vote Act, H.R. 11, 118th Congress.

STUDENTS PERSPECTIVE

Do we need to develop better methods for incorporating uncertainty into our geospatial analyses so there is motivation to develop better systems for tracking uncertainty in data? Or do we need to first have standardized metrics of uncertainty in order to develop these methods of handling uncertainty? **Who Should be leading this movement?**

MEASUREMENT

Promise or Illusion

”Put some quantitative meat on the bones of intuitive visual assessment.

Dozens of metrics, but many can often be used to make opposing cases.

Understand the metrics and assess what they can be used for

Compactness

Horn et al.	More valuable as a criterion for drawing districts than as proof of gerrymandering
Barnes & Solomon	Scores can be gamed, so standards must be precise and code must be open
Duchin & Tenner	Districting is built from administrative units, so consider graph-based measures

Polsby-Popper

Ratio of the area of region Ω to the area of the circle whose circumference is equal to the perimeter of Ω .

$$PP(\Omega) = 4\pi \frac{area(\Omega)}{perim(\Omega)^2}$$

Will this metric be
sensitive to perimeter?

Reock

Area of area of region Ω divided by the area of its minimum bounding circle, MBC

$$R(\Omega) = \frac{area(\Omega)}{area(MBC)}$$

Convex Hull Score

Area of area of region Ω divided by the area of its convex hull, CH

$$CHS(\Omega) = \frac{area(\Omega)}{area(CH)}$$

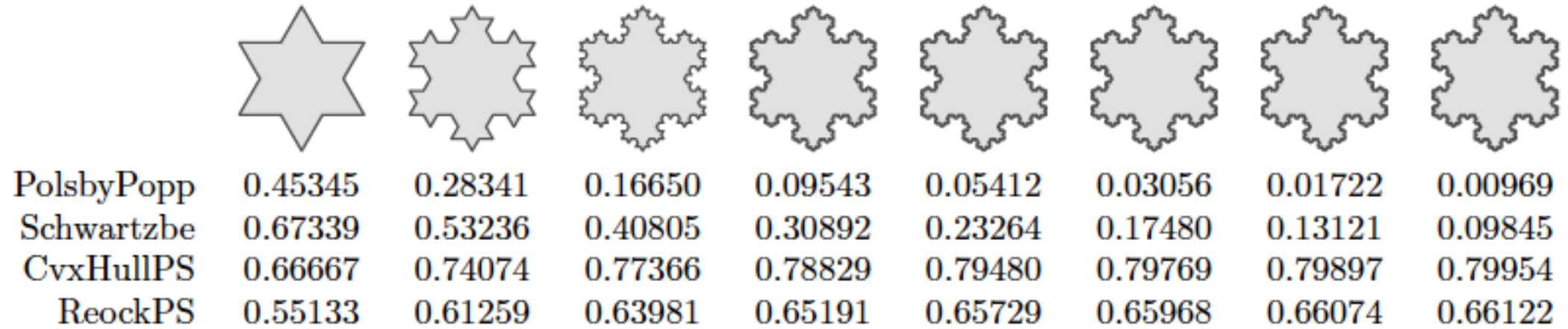


Figure 14: The Koch Snowflake (Von Koch 1904) shown for its first 8 levels of resolution (the 0th level is omitted). At each resolution both the shape and boundary of the snowflake are visually similar, especially at higher resolutions; however, the levels have markedly different scores. For each increase in resolution the Polsby–Popper score decreases by 77% and the Schwartzberg score by 33%. After initial increases, the Convex Hull and Reock scores stabilize.

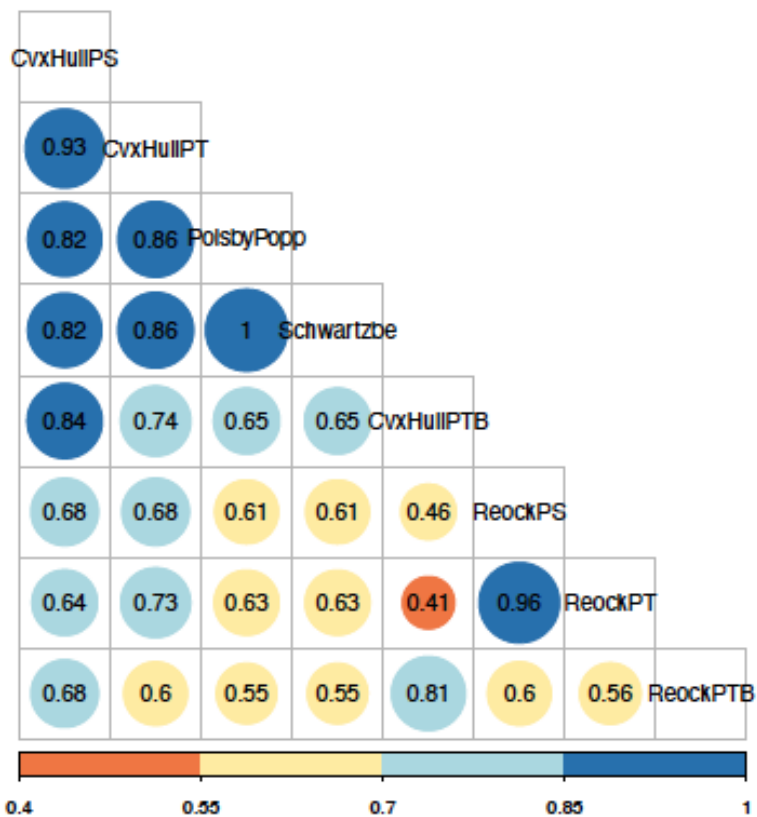


Figure 16: Correlations of rankings. Rankings of compactness scores for the 114th Congress at 1:500,000 resolution are compared against each other using Spearman’s rank correlation coefficient. A value of one indicates perfect agreement of relative rankings while a value of zero indicates no correlation.

STUDENTS PERSPECTIVE

It's interesting to me that we're using purely mathematical calculations to decide whether a district can produce a representative for an area and its people, when most of these measures are really just quantifying similarity. How can we extend compactness to include socioeconomic, demographic, and built-environment characteristics, basically calculating similarity of people and place rather than just polygon geometry? **Would a hybrid metric like this be more defensible than pure geometry or would it just open up new exploitable dimensions?**

Complicating Shape

Cumulative (maximum inscribed) circles method

- (1) Divide district into non-overlapping maximum inscribed circles
- (2) Euclidian distances from the circle centers to district centroid are standardized using attributes and topology
- (3) Standardized distances are aggregated to produce a single composite measure

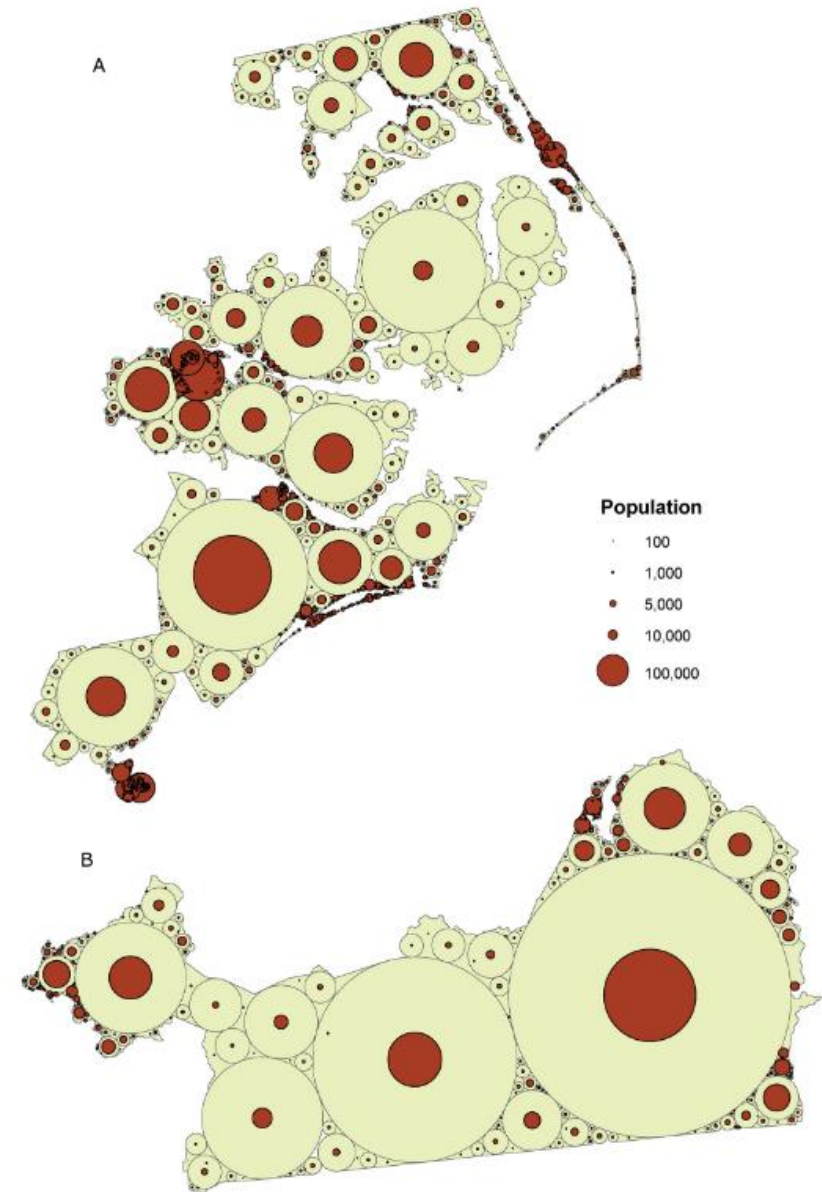


Figure 9. The (A) 3rd and (B) 10th Congressional districts in North Carolina. Each inscribed circle covers maximum population. Red circles are proportional to the populations covered by the inscribed circles.

Composition

Composition

What is the minority or party share in a district?

Wasted Votes

Votes for the winning candidate above 50%

Ensemble approach includes measure of shape, measure of composition, but is also fairly complex.

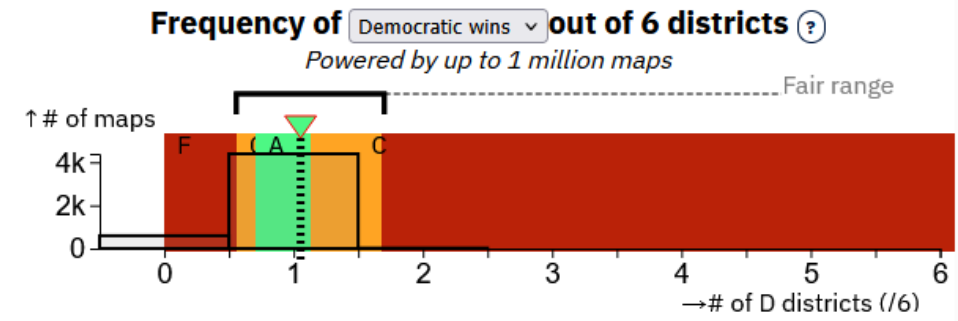
Ensemble Method

To generate a large set of alternative districting plans, we used the open-source package, [Gerrychain](#), that is commonly used in computational redistricting. We run the algorithm on precinct-level statewide files.

If the state did not have a reapportioned number of representatives, the last court-accepted plan was used as the starting partition. If the state had a different number of congressional districts than the previous cycle, a random partition was created and then optimized to pass the backsliding test and to have a reasonable number of cut edges (compactness) and county splits relative to the last court-accepted plan. This optimized partition closely matches the conditions in the last court-accepted plan and is used as the starting partition.

The ensemble constrains maps to have population equality within 5%, fewer or the same number of county splits and cut edges as the starting partition, and minority representation as dictated by the backsliding test. The algorithm is run for 1,000,000 steps which results in 1,000,000 alternative maps that strictly follow traditional redistricting criteria.

On this ensemble of alternative maps, we calculate the number of Democratic districts and the number of competitive districts using a partisan index that's the average of the most recent statewide elections for the U.S. President, U.S. Senate, and Governor. This gives us a distribution of the number of Democratic districts and number of competitive districts for maps that could have been drawn in the state that follow traditional criteria, but have no partisan intent.



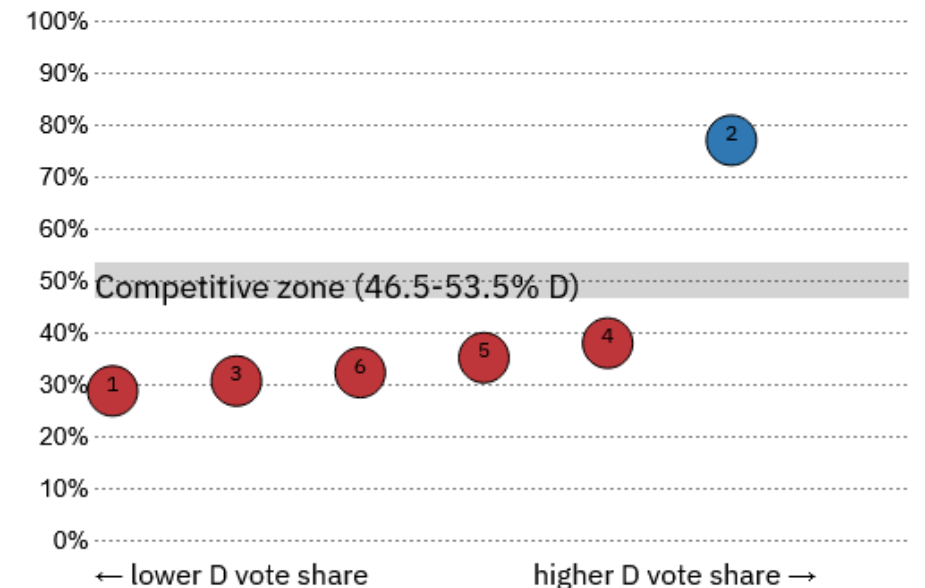
District by average partisan win percentage

 ?

Select a data point to see more info

Show as table Show Minimized chart Download as svg

D vote share



STUDENTS PERSPECTIVE

If compactness metrics are themselves unstable and open to manipulation through GIS workflows, **does the contemporary reliance on quantitative compactness standards simply shift political discretion into technical discretion?** More broadly, how should GIScience address the tension between reproducibility, transparency, and the inherently contingent nature of spatial measurement when quantitative metrics become embedded in legal and policy systems?

STUDENTS PERSPECTIVE

If you take the stance that gerrymandering is bad because it reduces competition in our political system and with it the generation of robust, new ideas to tackle our collective problems, GIS in this instance could be viewed as a tool that contributes negatively to society. **As GIScientists, should we be involved in mediating the application of GIS tools in broader society?**

BREAK

STUDENTS PERSPECTIVE

It seems that none of our readings discuss AWR sensitivity, and none reports the error this layer introduces. It feels like a hidden foundation nobody chooses to disclose.

The Zombie Dominion



ACTIVITY

Purpose:

To practice area-weighted reaggregation through spatial overlay, area weighting, and basic stats

Task:

- In groups of four zombies, work through area weighted reaggregation to plan you attack
- You should already be familiar with this geometry from you review of the Central Falls example
- The fastest and most accurate zombies will receive the GOLDEN BRAIN award and win a prize.

DOING SOME GERRYMANDERING

ACTIVITY

Purpose:

To familiarize you with the geographic components of Gerrymandering metrics and the analysis of Louisiana's congressional redistricting

Task:

- Reconstruct the overall workflow of a Louisiana gerrymandering study
- Use the **code** and **workflow components** provided to 'build' this analysis.
- Catalog the 'GIScience ideas' relevant to this task as you go

SETUP

Create a Local Copy of the Repository:

- Please work in pairs. At least one person in each pair should have the following already set up on their local machine: a GitHub account, Git/GitHub installed, and RStudio installed.
- Go to the course webpage. In the left-side menu bar, navigate to the Scale lesson, then open the Openshaw Effect lesson. In that lesson, you will find two links that we will use for today's activity: the GitHub repository that contains the activity materials, and the Google Sheet where we will record and compare our correlation coefficients.
- Open the GitHub repository linked from the lesson page. Click the Fork button in the upper-right corner of the GitHub page. This will create your own copy of the repository under your personal GitHub account. Make sure you are forking the repository named Learn-Gerrymandering.
- After you have forked the repository, clone your forked copy to your local machine. To do this, go to your forked repository on GitHub, click the green Code button, copy the repository URL, and use GitHub Desktop, the terminal, or another Git tool to clone it to a folder on your computer. Make sure you remember where you saved the cloned repository.

SETUP

Open and Run the Code:

- Once the repository has been cloned, navigate to the folder on your computer where you saved it. Inside the folder, find the R Project file. It should end in .Rproj. Open this file , not the .Rmd file or any other file first. Opening the .Rproj file will launch RStudio and automatically set the correct working directory to the folder containing the activity materials.
- After RStudio opens, use the Files pane to go to the data folder and open In-Class Activity.Rmd. This is the file we will work through during class.
- Start running the code from top to bottom. Read the instructions in the R Markdown file carefully as you go. Construct your workflow as you go.

STUDENTS PERSPECTIVE

While gerrymandering debates often center on electoral fairness, the follow-up question may lie in **what electoral representation actually delivers**: if the measure of fairness is equitable access to public goods, on what basis are public service districts drawn, and to what extent do they align with electoral districts?

Given that elected representatives hold the power to advocate for constituents but might bear no direct administrative responsibility for local service delivery, does this structural mismatch systematically disadvantage constituents?

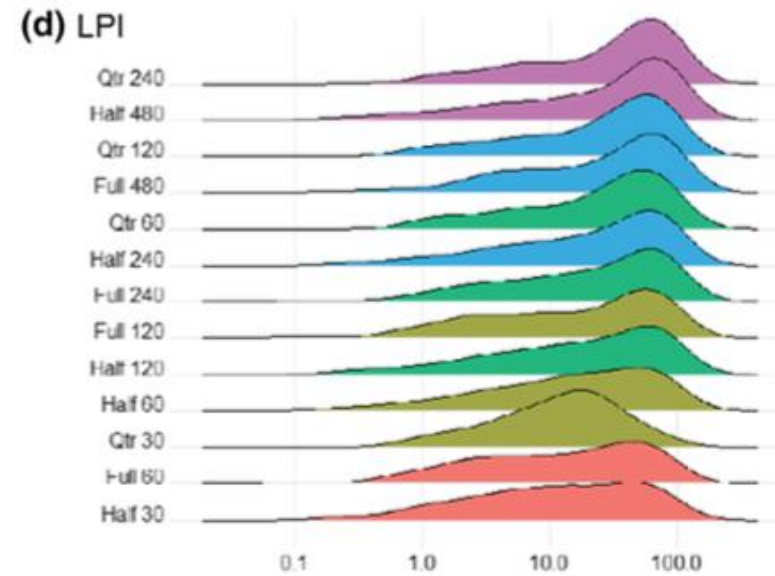
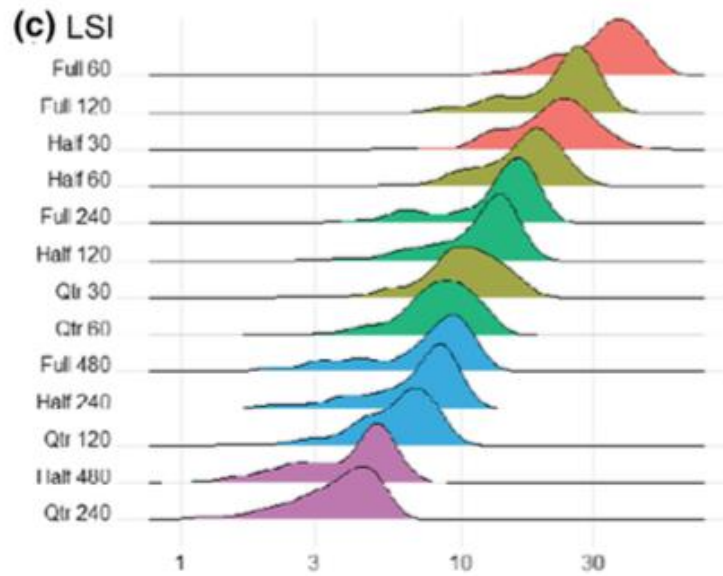
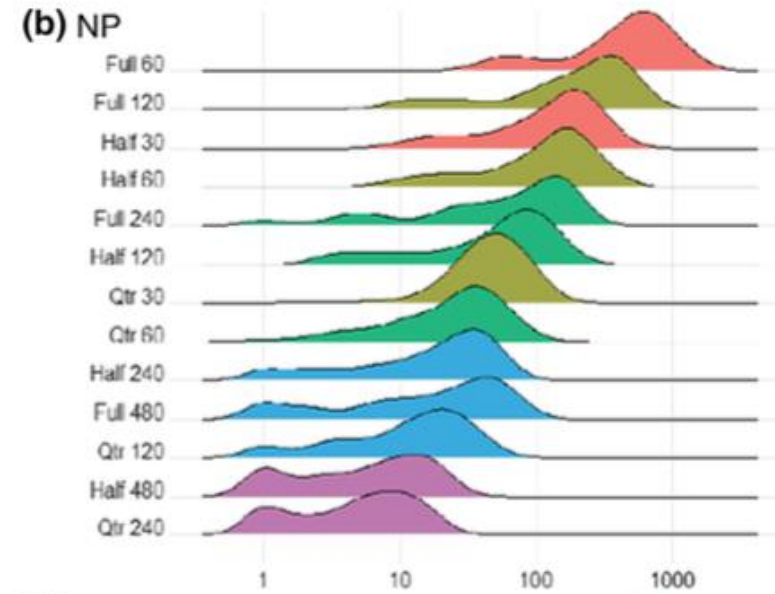
SCOPE

SCOPE

The dimensionless ratio between the range (extent) of a representation and the resolution (grain) of that representation

(a)

Grain km ² (res. m)	Extent		
	25 km ² (Qtr)	100 km ² (Half)	400 km ² (Full)
0.0009 (30 m)	B	A	
0.0036 (60 m)	C	B	A
0.0144 (120 m)	D	C	B
0.0576 (240 m)	E	D	C
0.2304 (480 m)		E	D



PRACTICAL ADVICE FROM DR FRAZIER

- FIRST** Researchers should set scope based on the phenomenon being studied, rather than relying solely on available data.
- SECOND** Studies should report both grain and extent quantitatively so that scope can be calculated and compared across studies.
- THIRD** Researchers should undertake comparisons and replications based on scope.

STUDENT PERSPECTIVE

Should spatial objects have the ability to preserve native resolutions after steps like aggregation? What implications would this have for replicability in research?